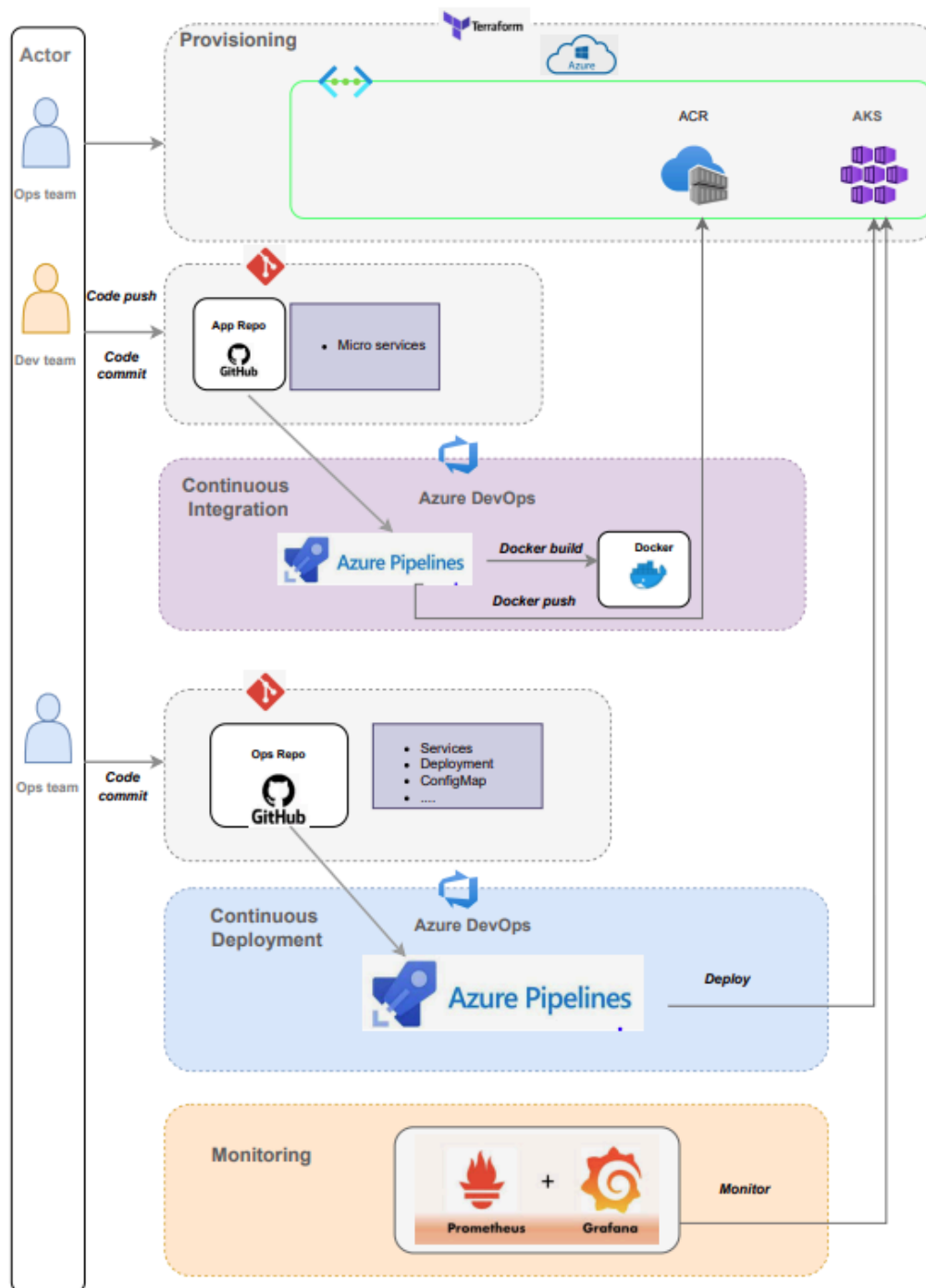


Setting up a CI/CD Pipeline and deploying applications on Azure AKS



I. Provision Azure Resources

Prerequisites for Using Terraform with Azure

1. Terraform Installed

- Install Terraform
 - Download Terraform

```
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
```

Add the official HashiCorp repo

```
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com $(lsb_release -cs) main" | \
sudo tee /etc/apt/sources.list.d/hashicorp.list
```

Update and install

```
sudo apt update
```

```
sudo apt install terraform -y
```

```
Processing triggers for man-db (5.10-1) ...
vuongnguyenv@VNNOT02236:~$ curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
o "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com focal main" | sudo tee /etc/apt/sources.list.d/hashicorp.list

sudo apt update
sudo apt install terraform -y

terraform -v # verify installation
vuongnguyenv@VNNOT02236:~$
vuongnguyenv@VNNOT02236:~$ echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com focal main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com focal main
```

```
list.d/kubernetes.list.d/
vuongnguyenv@VNNOT02236:~$ sudo apt install terraform -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 74 not upgraded.
Need to get 28.4 MB of archives.
After this operation, 93.6 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com focal/main amd64 terraform amd64 1.12.1-1 [28.4 MB]
Fetched 28.4 MB in 2s (14.2 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 48325 files and directories currently installed.)
Preparing to unpack .../terraform_1.12.1-1_amd64.deb ...
Unpacking terraform (1.12.1-1) ...
Setting up terraform (1.12.1-1) ...
```

2. Azure CLI Installed and Authenticated

- Install Azure CLI

```

vuongnguyenv@VNNOT02236:~$ sudo apt update
Install required dependencies
sudo apt install -y ca-certificates curl apt-transport-https lsb-release gnupg

# 3. Download and install the Microsoft signing key
curl -sL https://packages.microsoft.com/keys/microsoft.asc | \
  gpg --dearmor | \
  sudo tee /etc/apt/trusted.gpg.d/microsoft.gpg > /dev/null

# 4. Add the Azure CLI software repository
AZ_REPO=$(lsb_release -cs)
echo "deb [arch=amd64] https://packages.microsoft.com/repos/azure-cli/ $AZ_REPO main" | \
  sudo tee /etc/apt/sources.list.d/azure-cli.list

# 5. Update package lists again and install Azure CLI
sudo apt update
sudo apt install -y azure-cli

# 6. Check if it worked
az version

```

```

list.d/rubermetes.list:2
vuongnguyenv@VNNOT02236:~$ sudo apt install -y azure-cli
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  azure-cli
0 upgraded, 1 newly installed, 0 to remove and 69 not upgraded.
Need to get 54.4 MB of archives.
After this operation, 710 MB of additional disk space will be used.
Get:1 https://packages.microsoft.com/repos/azure-cli focal/main amd64 azure-cli amd64 2.72.0-1~focal [54.4 MB]
Fetched 54.4 MB in 7s (8165 kB/s)
Selecting previously unselected package azure-cli.
(Reading database ... 48328 files and directories currently installed.)
Preparing to unpack .../azure-cli_2.72.0-1~focal_amd64.deb ...
Unpacking azure-cli (2.72.0-1~focal) ...
Setting up azure-cli (2.72.0-1~focal) ...

```

• Authenticate to Azure

```

vuongnguyenv@VNNOT02236:~$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in
the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device
-code'.
gio: https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize?client_id=04b07795-8ddb-461a-bbee-02f9e1bf7b46&response_ty
pe=code&redirect_uri=http%3A%2F%2Flocalhost%3A45503&scope=https%3A%2F%2Fmanagement.core.windows.net%2F%2F.default-offline_access+open
id+profile&state=tKEdBp0TrFunXiHc&code_challenge=DjZWlvJlgxN30zsBD7WkyafQrUsF7RBWhq1xGehTckI&code_challenge_method=S256&nonce=06fefc7
516bd5558b20c3fdd141966bbc459896f55992d9e3cbe5dc975177b4f&client_info=1&claims=%7B%22access_token%22%3A+%7B%22xms_cc%22%3A+%7B%22valu
es%22%3A+%5B%22CP1%22%5D%7D%7D&prompt=select_account: Operation not supported

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   demo-subscription      892caa5d-118e-4e71-a834-8af725f0a94e   Default Directory

The default is marked with an *; the default tenant is 'Default Directory' and subscription is 'demo-subscription' (892caa5d-118e-4e7
1-a834-8af725f0a94e).

Select a subscription and tenant (Type a number or Enter for no changes): demo-subscription
Invalid selection.
Select a subscription and tenant (Type a number or Enter for no changes):

Tenant: Default Directory
Subscription: demo-subscription (892caa5d-118e-4e71-a834-8af725f0a94e)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its con

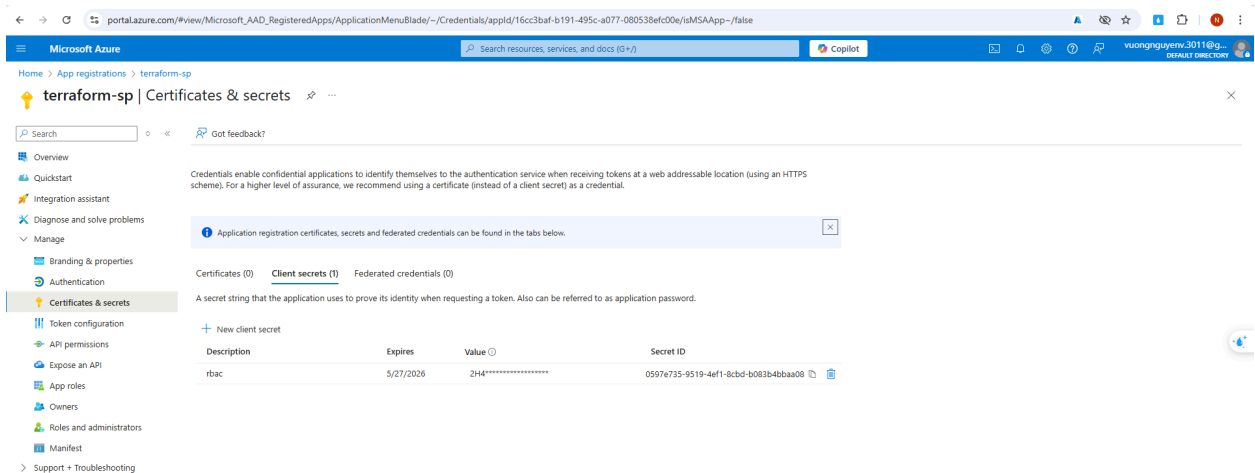
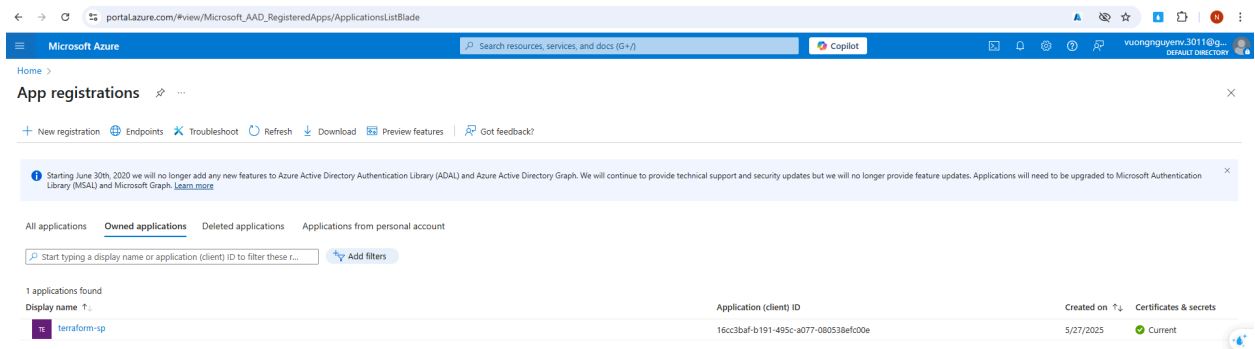
```

- **Create service principle**

```
psFordev/terraform-demo/terraformform$ az ad sp create-for-rbac --name terraform-sp --role "Owner" --scopes /subscriptions/892c  
aa5d-118e-4e71-a834-8af725f0a94e  
Creating 'Owner' role assignment under scope '/subscriptions/892caa5d-118e-4e71-a834-8af725f0a94e'  
The output includes credentials that you must protect. Be sure that you do not include these credentials in your code or check the cr  
edentials into your source control. For more information, see https://aka.ms/azadsp-cli  
{  
  "appId": "16cc3baf-b191-495c-a077-080538efc00e",  
  "displayName": "terraform-sp",  
  "password": "2H10Q1W1G1P1L1A1K1ZANXVLM1ZF1FN1J1U1B1",  
  "tenant": "d2ae8cde-adca-4dcc-a221-cfc3d7f44737"
```

- **Terraform Azure Provider Setup**

Access to **Azure Portal** → **App registrations**



[Delete](#)
[Endpoints](#)
[Preview features](#)

ⓘ Got a second? We would love your feedback on Microsoft identity platform (previously Azure AD for developer). →

^ Essentials

Display name	: terraform-sp	Client credentials	: 0 certificate_1 secret
Application (client) ID	: 16cc3baf-b191-495c-a077-080538efc00e	Redirect URIs	: Add a Redirect URI
Object ID	: 1cc5c419-b6b5-4d06-a9a5-f91cb8894e70	Application ID URI	: Add an Application ID URI
Directory (tenant) ID	: d2ae8cde-adca-4dcc-a221-cfc3d7f44737	Managed application in L...	: terraform-sp

Supported account types : [My organization only](#)

ⓘ Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

[Get Started](#)
[Documentation](#)

Input value to provider in main.tf

```

terraform-demo > terraform > main.tf
1  terraform {
2    required_providers {
3      azurerm = {
4        source  = "hashicorp/azurerm"
5        version = "~> 3.0"
6      }
7    }
8  }
9
10 provider "azurerm" {
11   tenant_id = "d2ae8cde-adca-4dcc-a221-cfc3d7f44737"
12   client_id = "16cc3baf-b191-495c-a077-080538efc00e"
13   client_secret = "2H46q...wzqneq4p...E0n3XNtHE3r...g50...9j...7"
14   subscription_id = "892caa5d-118e-4e71-a834-8af725f0a94e"
15   features {}
16 }
17
  
```

Run: **terraform init** to initialize

```

vuongnguyenv@VNNOT02236:/mnt/c/Users/vuongnguyenv/Documents/ReMS/DevopsFordev/terraform-demo/terraform$ terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/azurerm from the dependency lock file
- Using previously-installed hashicorp/azurerm v3.117.1

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
  
```

Run: **terraform validate** to validate scripts

```
vuongnguyenv@VNNOT02236:/mnt/c/Users/vuongnguyenv/Documents/ReMS/DevopsFordev/terraform-demo/terraform$ terraform validate
Success! The configuration is valid.
```

Run: **terraform plan -var-file=terraform.tfvars -out=terraform.plan** to generate an execution plan with custom variable values and save the plan to a file.

```
vuongnguyenv@VNNOT02236:/mnt/c/Users/vuongnguyenv/Documents/ReMS/DevopsFordev/terraform-demo/terraform$ terraform plan -var-file=terraform.tfvars -out=terraform.plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# module.acr.azure_rm_container_registry.main will be created
+ resource "azure_rm_container_registry" "main" {
  + admin_enabled           = true
  + admin_password          = (sensitive value)
  + admin_username          = (known after apply)
  + encryption              = (known after apply)
  + export_policy_enabled   = true
  + id                      = (known after apply)
  + location                 = "southeastasia"
  + login_server             = (known after apply)
  + name                    = "msappracticeacr"
  + network_rule_bypass_option = "AzureServices"
  + network_rule_set         = (known after apply)
  + public_network_access_enabled = true
  + resource_group_name      = "sd4413-practical-devops-resource-group"
  + retention_policy         = (known after apply)
  + sku                      = "Basic"
  + trust_policy             = (known after apply)
  + zone_redundancy_enabled = false
}

# module.aks.azure_rm_kubernetes_cluster.main will be created
+ resource "azure_rm_kubernetes_cluster" "main" {
  + api_server_authorized_ip_ranges = (known after apply)
  + current_kubernetes_version      = (known after apply)
  + dns_prefix                       = "aksdns"
  + fqdn                            = (known after apply)
  + name                            = (known after apply)
  + principal_id                     = (known after apply)
  + principal_type                   = (known after apply)
  + role_definition_id               = (known after apply)
  + role_definition_name             = "AcrPush"
  + scope                           = "/subscriptions/892caa5d-118e-4e71-a834-8af725f0a94e/resourceGroups/sd4413-practical-devops-resource-group/providers/Microsoft.ContainerRegistry/registries/msappracticeacr"
  + skip_service_principal_aad_check = (known after apply)
}

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ aks_kubeconfig = (sensitive value)

Warning: Argument is deprecated

  with module.aks.azure_rm_kubernetes_cluster.main,
  on modules/aks/main.tf line 22, in resource "azure_rm_kubernetes_cluster" "main":
  22:     docker_bridge_cidr = "172.17.0.1/16"

'docker_bridge_cidr' has been deprecated as the API no longer
supports it and will be removed in version 4.0 of the
provider.

Saved the plan to: terraform.plan

To perform exactly these actions, run the following command to
apply:
  terraform apply "terraform.plan"
```

Run: terraform apply terraform.plan

```
vuongnguyenv@VNNOT02236: /mnt/c/Users/vuongnguyenv/Documents/ReMS/DevopsFordev/terraform-demo/terraform$ terraform apply terraform.plan
module.aks.azure_rm_kubernetes_cluster.main: Creating...
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [00m10s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [00m20s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [00m30s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [00m40s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [00m50s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m00s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m10s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m20s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m30s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m40s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [01m50s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m00s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m10s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m20s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m30s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m40s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [02m50s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [03m00s elapsed]
module.aks.azure_rm_kubernetes_cluster.main: Still creating... [03m10s elapsed]
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:
acr_login_server = "msappracticeacr.azurecr.io"
aks_kubeconfig = <sensitive>
```

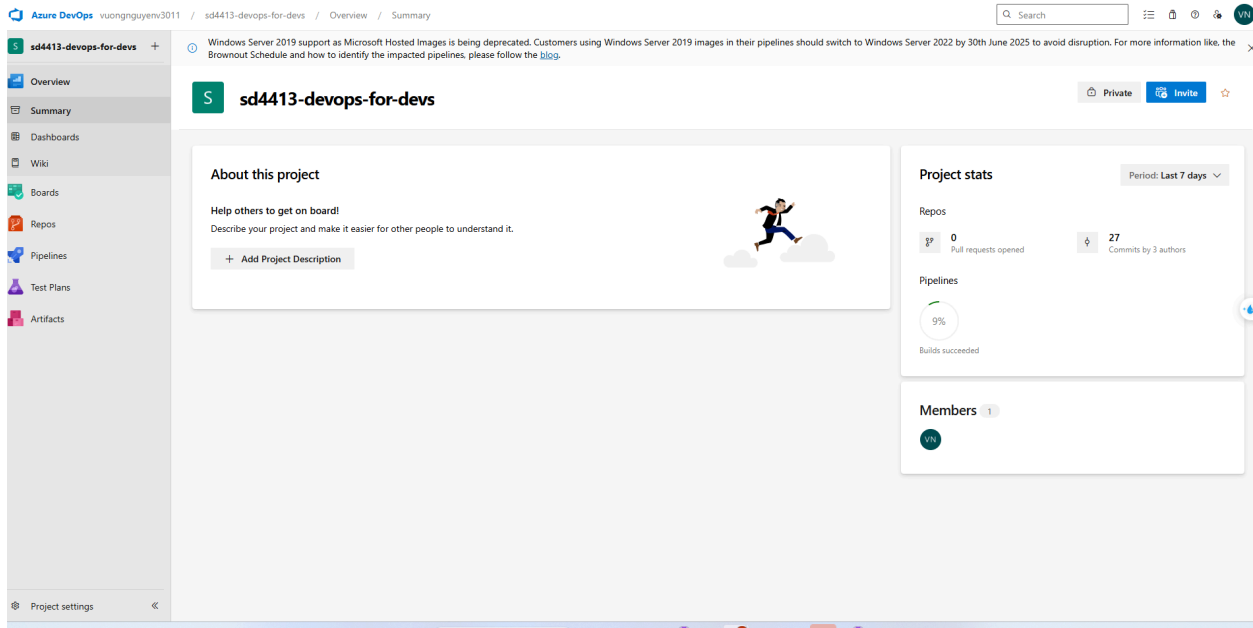
- Recheck on Azure Portal

The screenshot shows the Azure Portal interface for the resource group 'sd4413-practical-devops-resource-group'. The 'Resources' tab is active, showing a list of resources. The table has columns for Name, Type, and Location. The resources listed are:

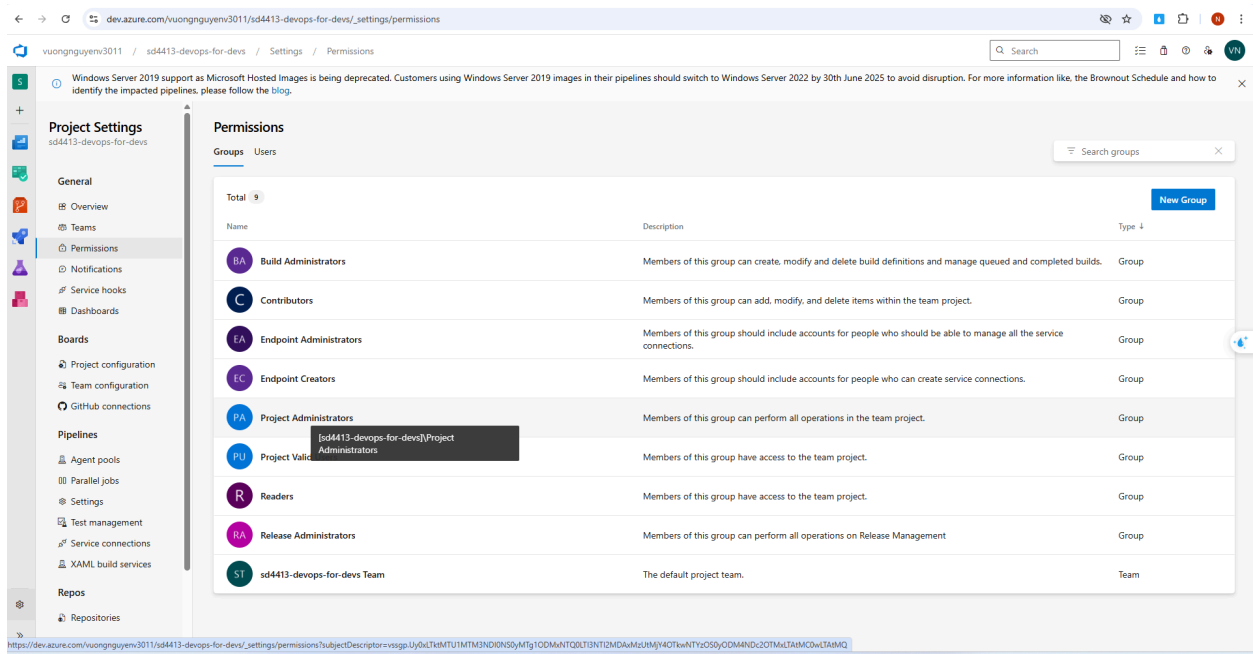
Name	Type	Location
msa-aks-cluster	Kubernetes service	Southeast Asia
msa-vnet	Virtual network	Southeast Asia
msappracticeacr	Container registry	Southeast Asia

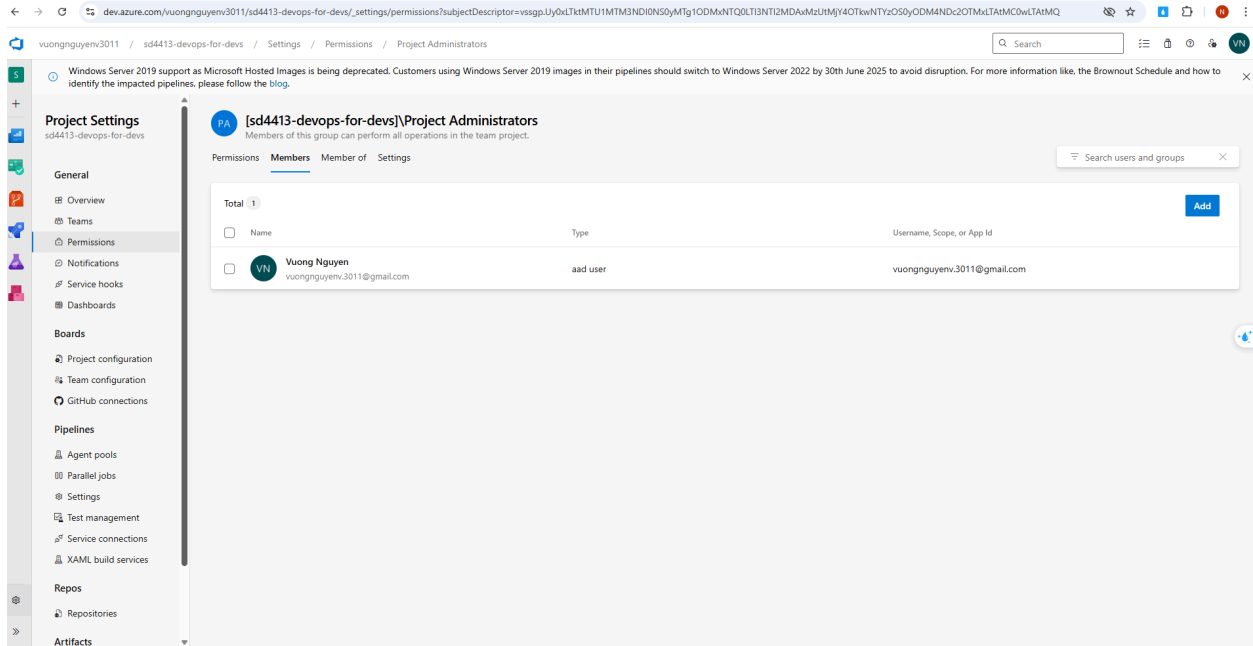
II. Setup Azure DevOps pipeline for CI/CD

- Create new project in portal



• Grant permission to account





- Setup agent with **Self-hosted agent** To Run CI/CD Pipelines

Windows

macOS

Linux

x64

arm64

arm

System prerequisites

Configure your account

Configure your account by following the steps outlined here.

Download the agent

[Download](#)

Create the agent

```
~/ $ mkdir myagent && cd myagent  
~/myagent$ tar zxvf ~/Downloads/vsts-agent-linux-x64-4.255.0.tar.gz
```

Configure the agent [Detailed instructions](#)

```
~/myagent$ ./config.sh
```

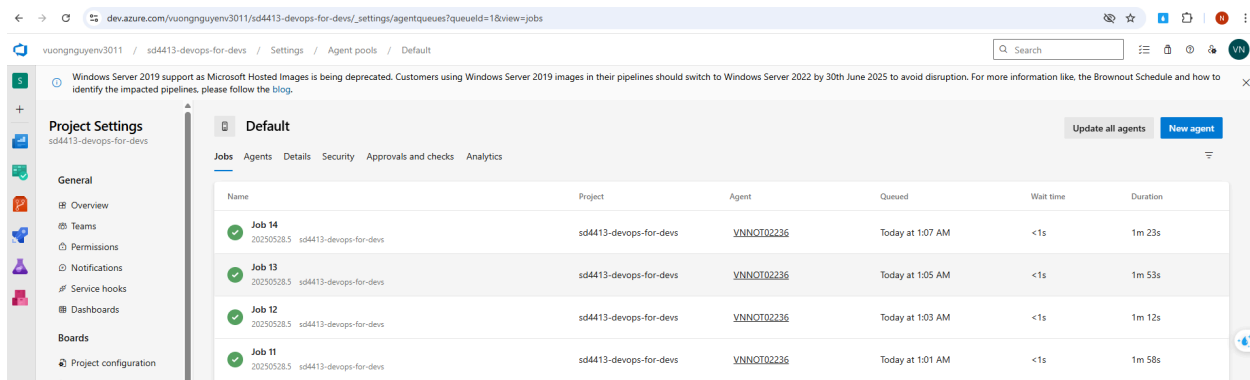
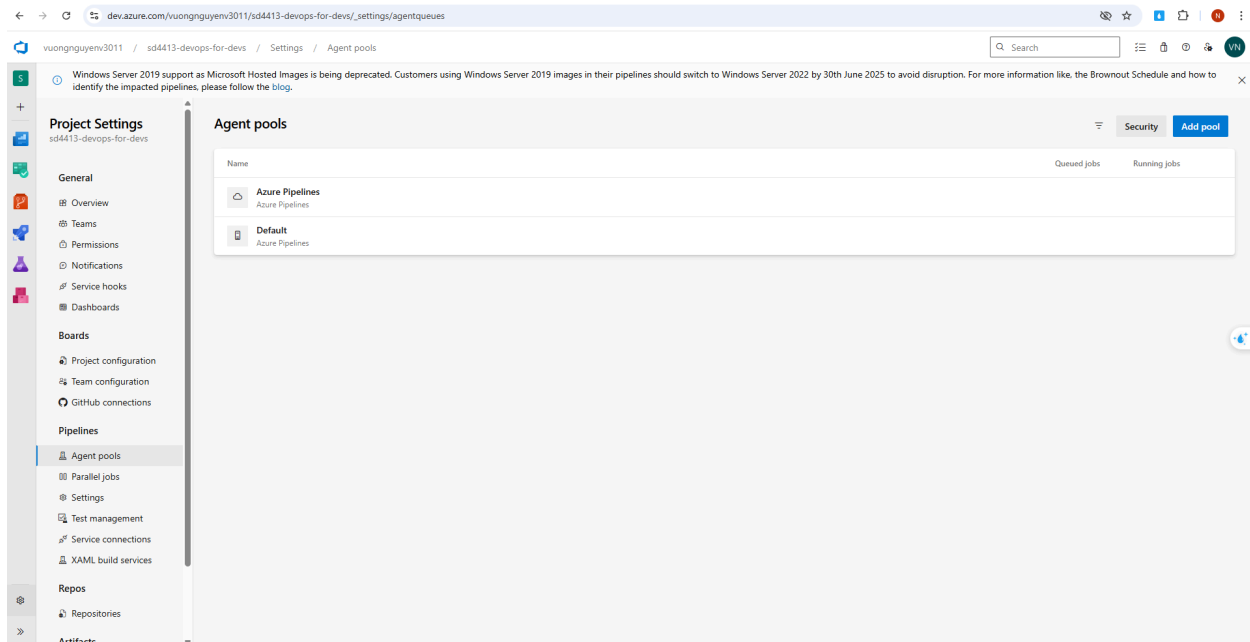
Optionally run the agent interactively

If you didn't run as a service above:

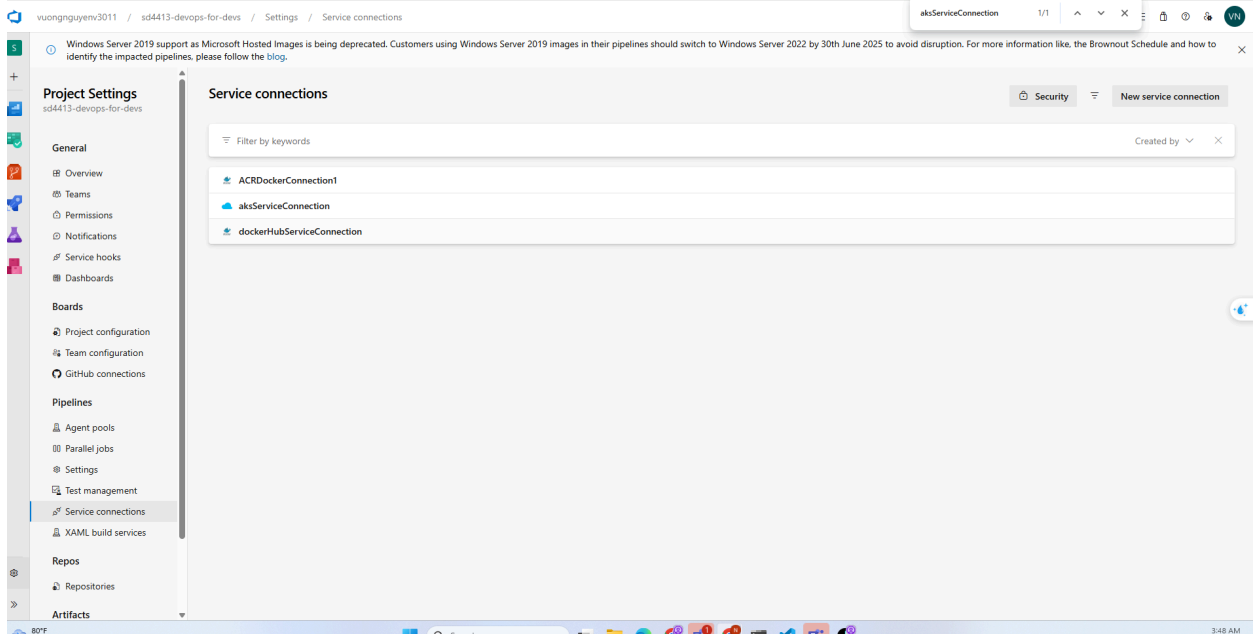
```
~/myagent$ ./run.sh
```

That's it!

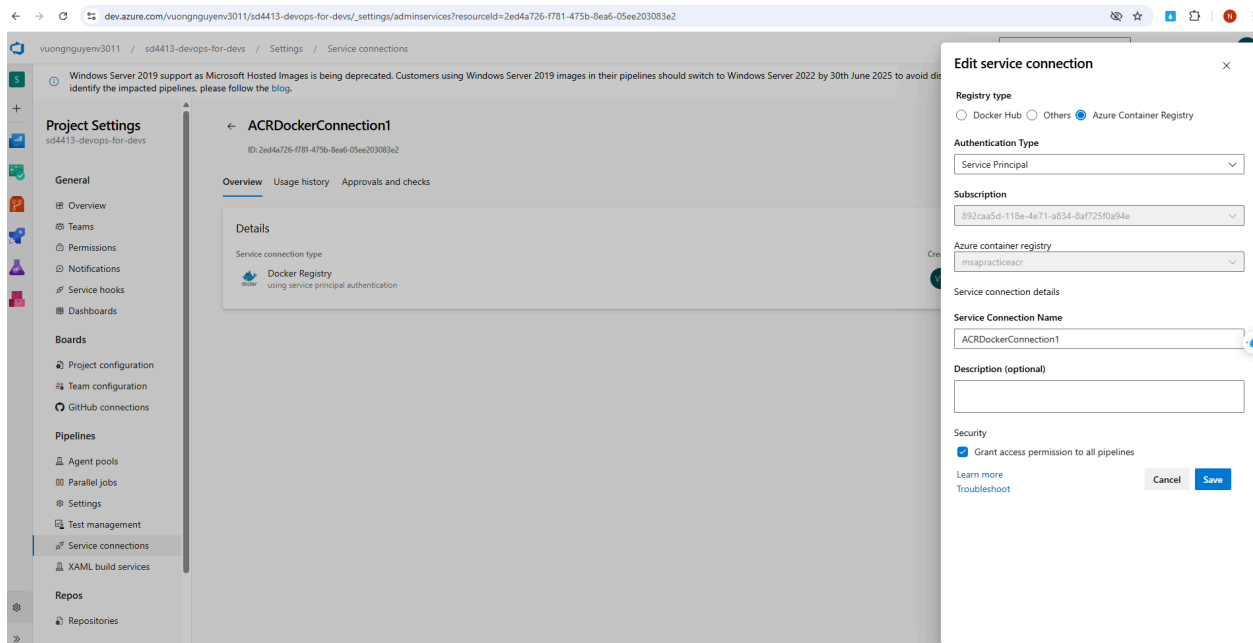
[More Information](#)



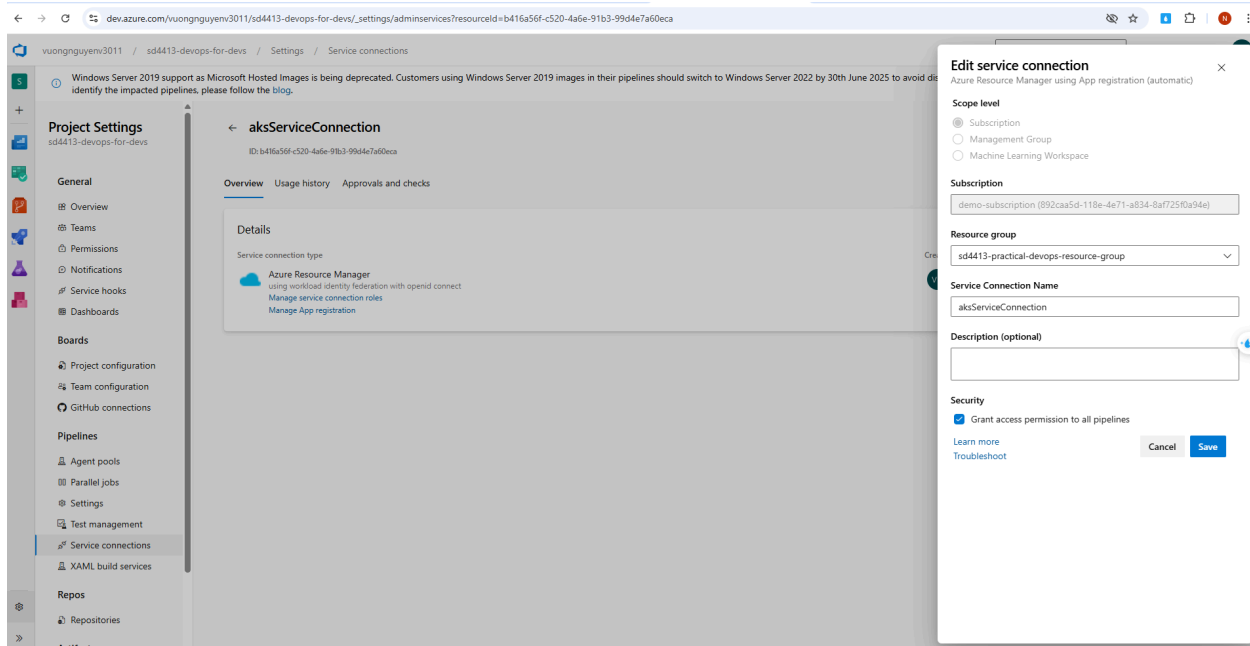
- **Setup service connections**



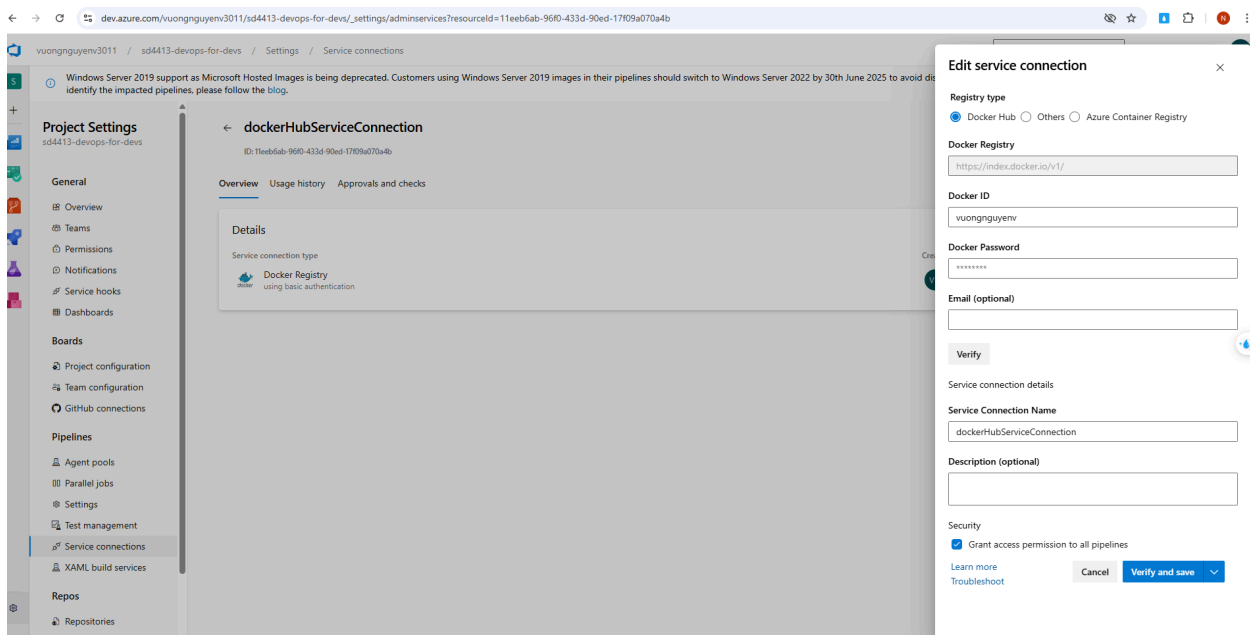
- Setup **ACRDockerConnection1** to push **Azure Container Register**



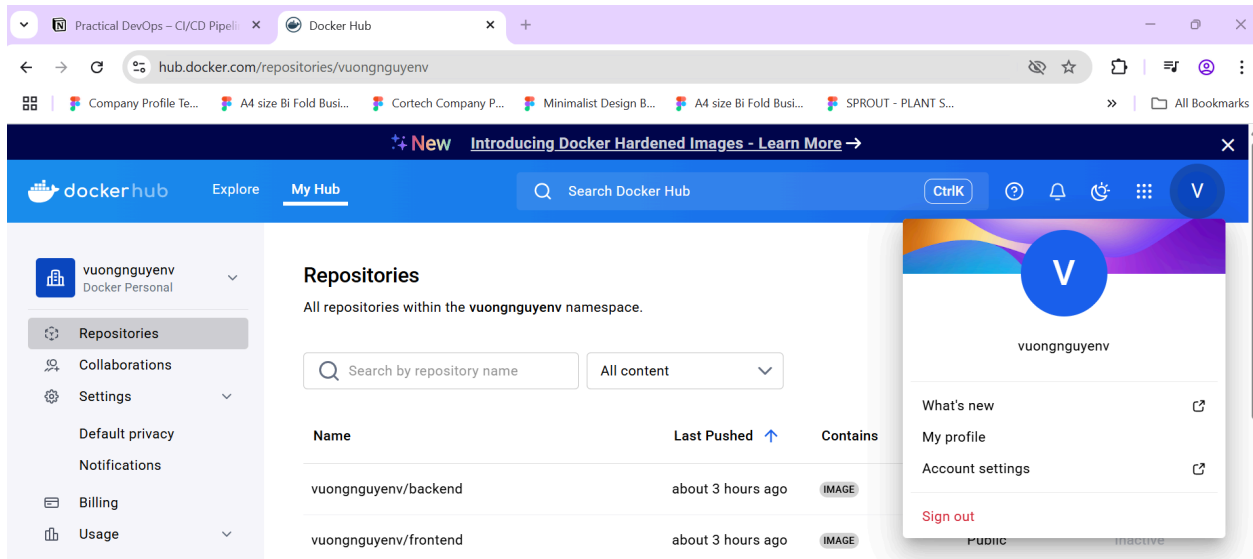
- Setup **aksServiceConnection** to Deploy Frontend/Backend/DB to AKS to Docker Hub



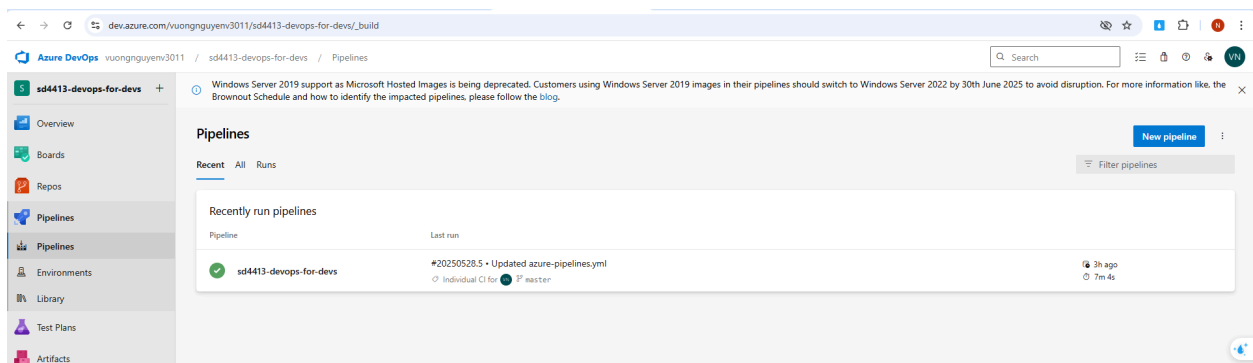
- Setup **dockerHubServiceConnection** to Build & Push to Docker Hub



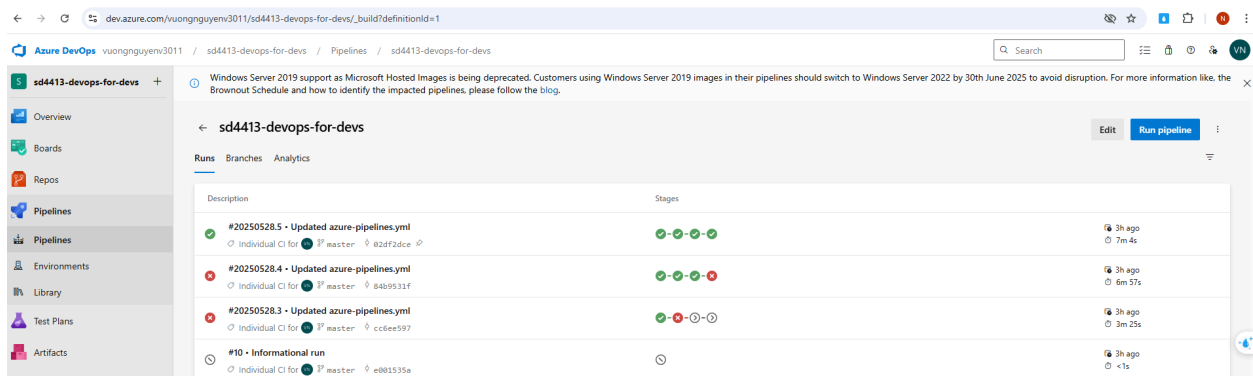
With Docker Id is username on page <https://hub.docker.com/> after logged in.



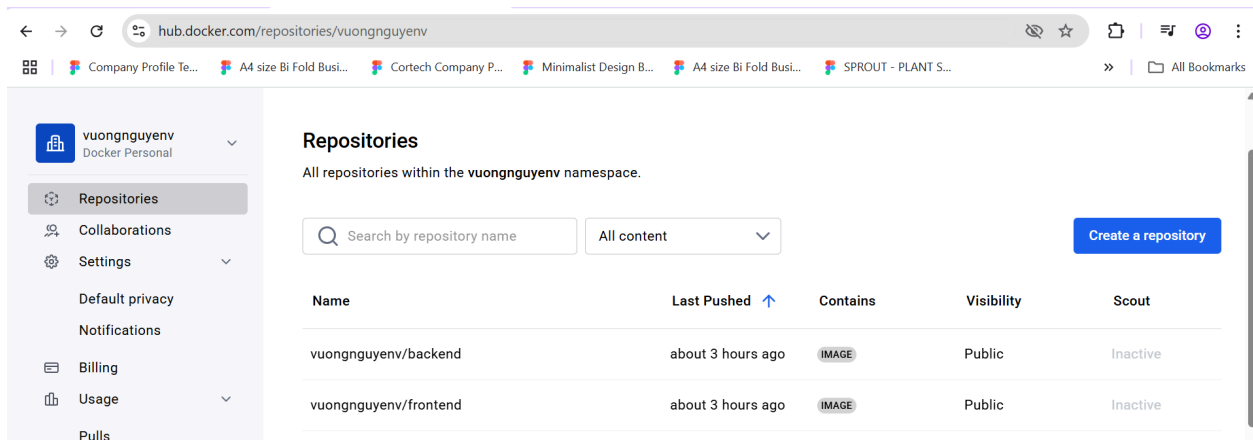
- After setup pipelines like this:



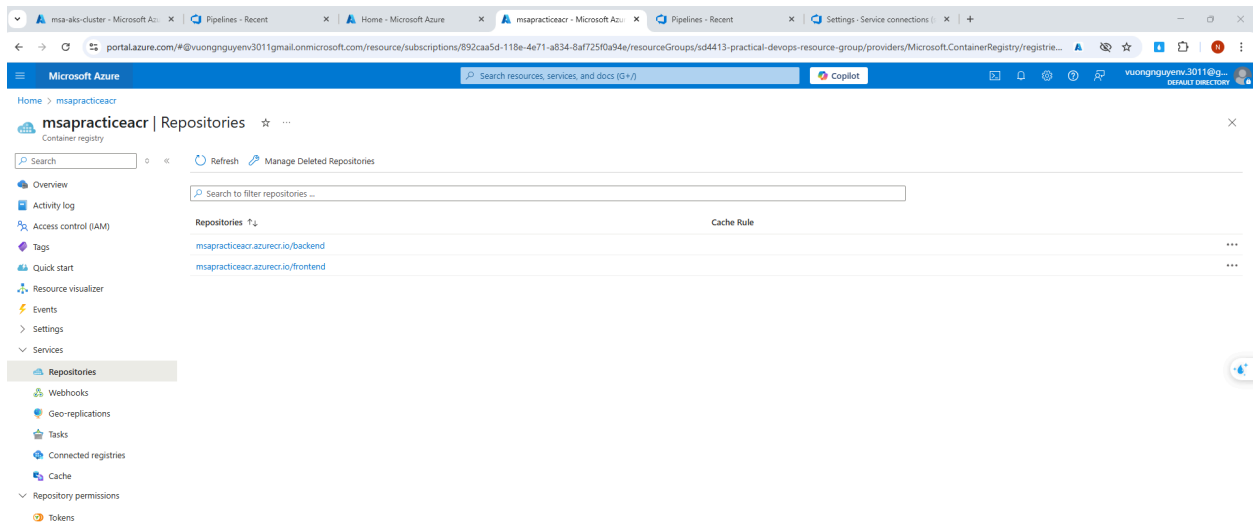
- Check status pipelines



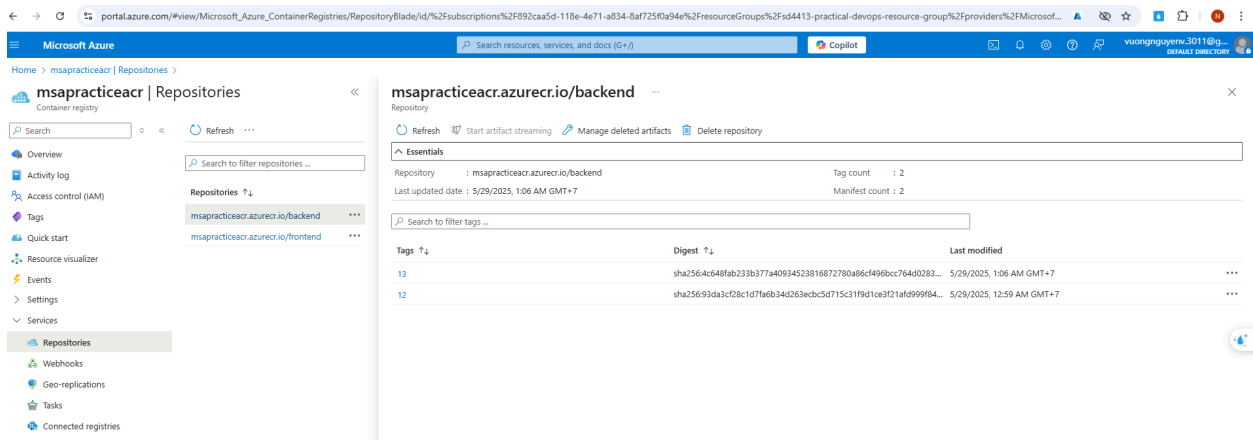
- Verify repositories on docker hub



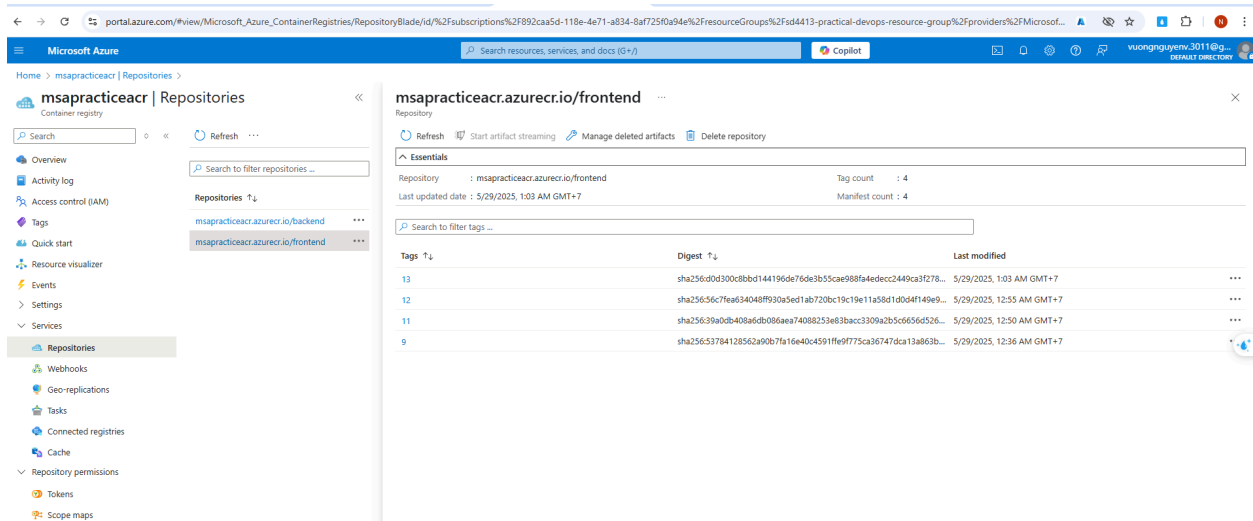
- Verify repositories on Azure Container Registry



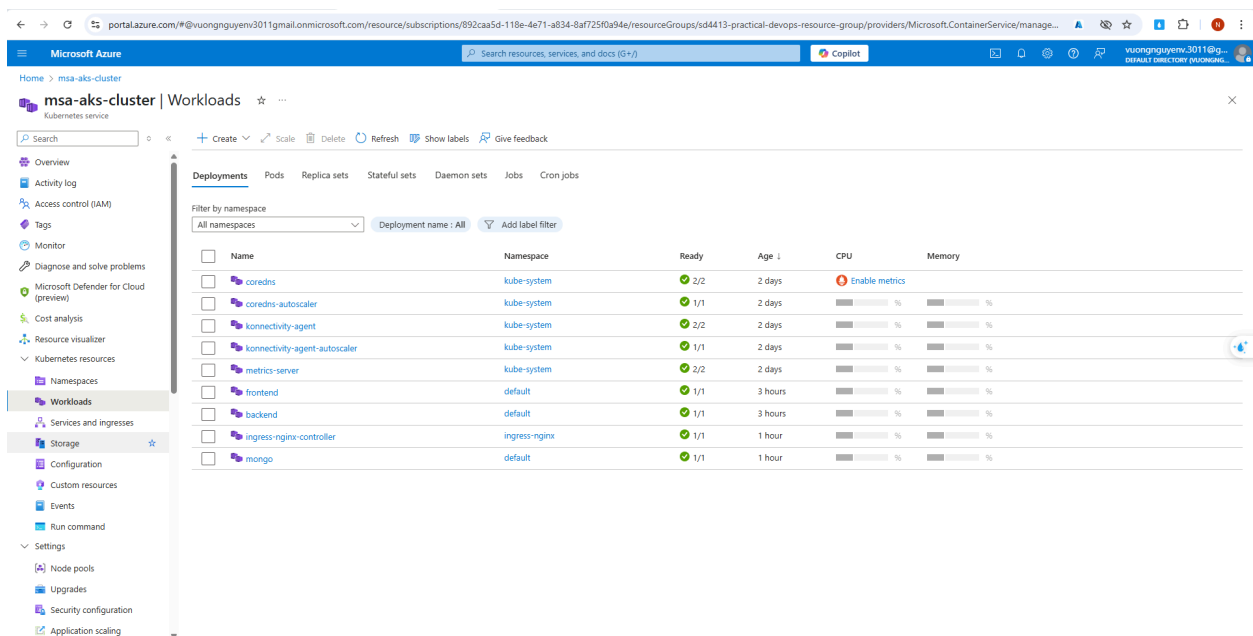
- Verify Backend repository on Azure Container Registry



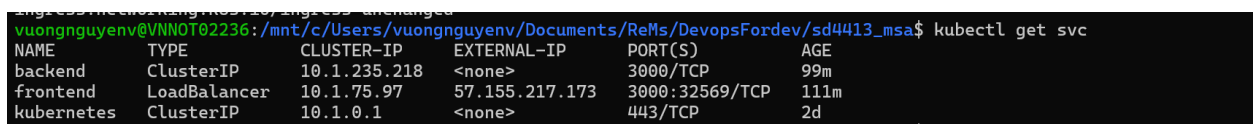
- Verify Fontend repository on Azure Container Registry



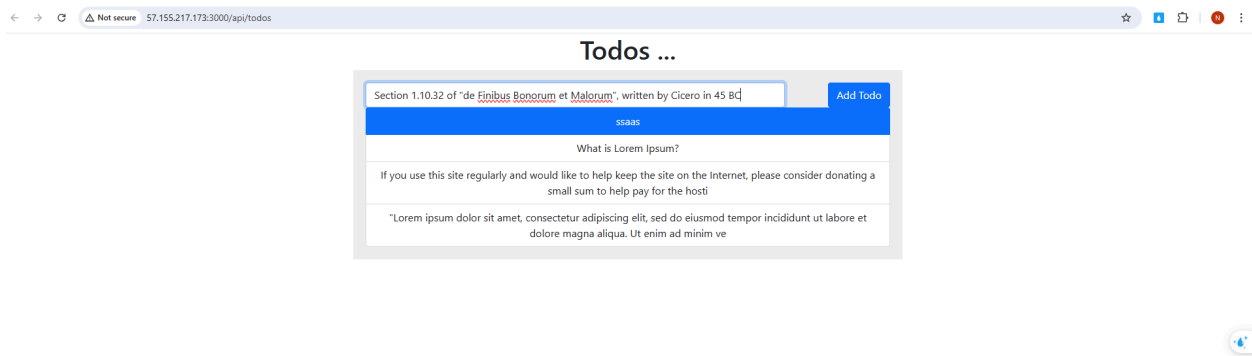
- Verify Backend / Front end/ Db cluster in Azure Kubernetes Service



- Get extenal-ip on frontend using Load Balancer use to expose



- Access url: <http://57.155.217.173:3000/> from results above.



III. Monitoring

Install Grafana and Prometheus by helm command

```
helm repo add prometheus-community https://prometheus-community.github.io/
```

```
helm repo add stable https://charts.helm.sh/stable
```

```
helm repo update
```

```
helm pull prometheus-community/kube-prometheus-stack --version 48.1.1
```

```
tar -xzf kube-prometheus-stack-48.1.1.tgz
```

- Override the "prometheus-community/kube-prometheus-stack"

```
cp kube-prometheus-stack/values.yaml ./kube-prometheus-stack-values.yaml
```

- Use Vs Code is editor to edit kube-prometheus-stack-values.yaml file

```
vuongnguyenv@VNNOT02236:~$ cp kube-prometheus-stack/values.yaml ./kube-prometheus-stack-values.yaml
vuongnguyenv@VNNOT02236:~$ dir
kube-prometheus-stack  kube-prometheus-stack-48.1.1.tgz  kube-prometheus-stack-values.yaml  snap
vuongnguyenv@VNNOT02236:~$ code kube-prometheus-stack-values.yaml
vuongnguyenv@VNNOT02236:~$ code kube-prometheus-stack-values.yaml
```

With content and save the file.

podMonitorSelectorNilUsesHelmValues: false

serviceMonitorSelectorNilUsesHelmValues: false

ruleSelectorNilUsesHelmValues: false

Run: kubectl get ns to get all name spacing

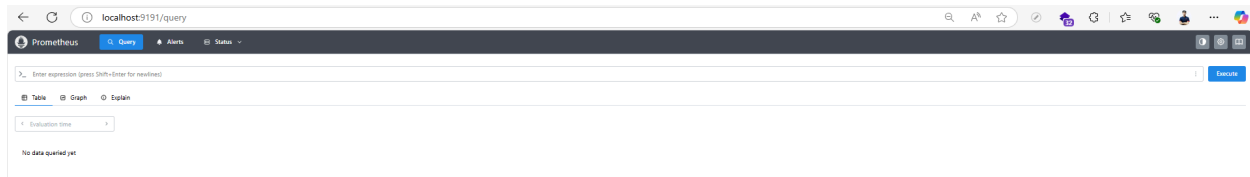
```
6m57s
vuongnguyenv@VNNOT02236:~$ kubectl get ns
NAME                STATUS    AGE
default             Active   2d20h
ingress-nginx       Active   19h
kube-node-lease     Active   2d20h
kube-public         Active   2d20h
kube-system         Active   2d20h
monitoring          Active   18m
```

- Currently backend, frontend, mongo services are running with name spacing is default

```
vuongnguyenv@VNNOT02236:~$ kubectl get svc -n default
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)
backend             ClusterIP     10.1.235.218  <none>          3000/TCP
frontend            LoadBalancer 10.1.75.97    57.155.217.173 3000:32569
kubernetes          ClusterIP     10.1.0.1      <none>          443/TCP
mongo               ClusterIP     10.1.244.30   <none>          27017/TCP
prometheus-alertmanager ClusterIP     10.1.66.108   <none>          9093/TCP
prometheus-alertmanager-headless ClusterIP     None          <none>          9093/TCP
prometheus-kube-state-metrics ClusterIP     10.1.233.193  <none>          8080/TCP
prometheus-prometheus-node-exporter ClusterIP     10.1.50.34    <none>          9100/TCP
prometheus-prometheus-pushgateway ClusterIP     10.1.242.242  <none>          9091/TCP
prometheus-server   ClusterIP     10.1.170.207  <none>          80/TCP
```

- **Access Prometheus by using port-forward**

```
vuongnguyenv@VNNOT02236:~$ kubectl port-forward -n monitoring svc/prometheus-operated 9191:9090
Forwarding from 127.0.0.1:9191 -> 9090
Forwarding from [::1]:9191 -> 9090
Handling connection for 9191
Handling connection for 9191
Handling connection for 9191
Handling connection for 9191
Handling connection for 9191
```

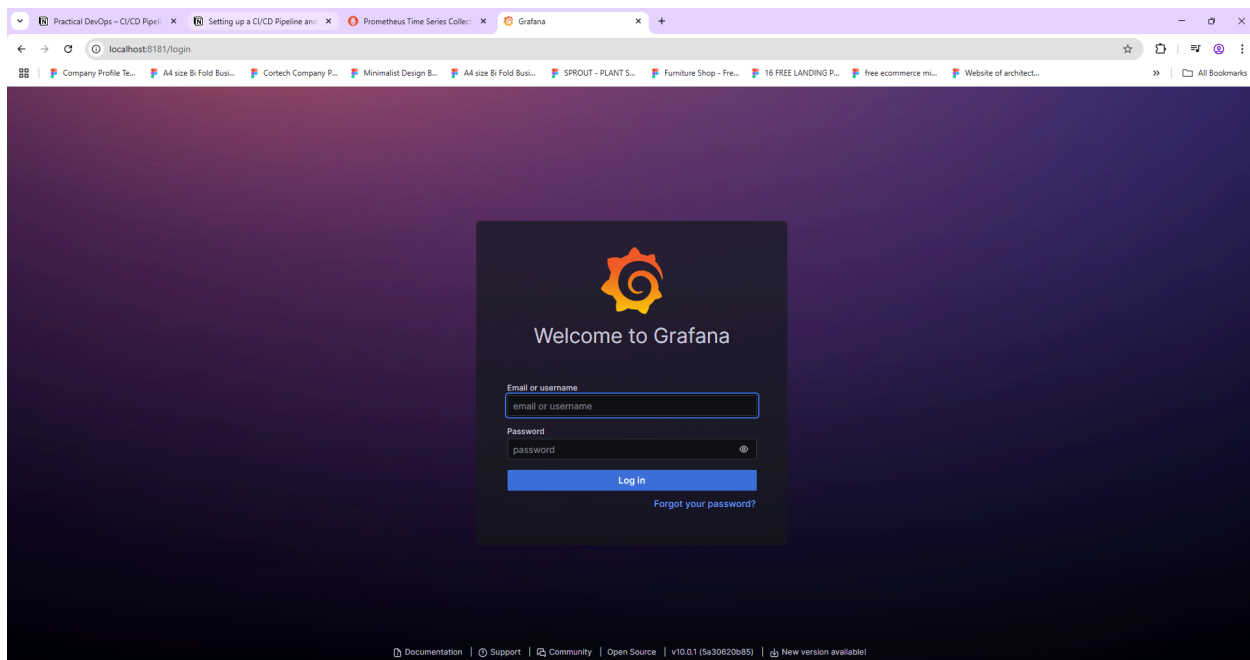


- **Access Grafana by using port-forward**

```

vuongnguyenv@VNNOT02236:~$ kubectl port-forward -n monitoring svc/prometheus-grafana-stack 8181:80
Forwarding from 127.0.0.1:8181 -> 3000
Forwarding from [::1]:8181 -> 3000
Handling connection for 8181
Handling connection for 8181
Handling connection for 8181
Handling connection for 8181
Handling connection for 8181
Handling connection for 8181
Handling connection for 8181

```



- **To login on page: Run get password and user name default is admin**

`kubectl get secret -n monitoring prom-operator-grafana -o jsonpath="{.data.admin-}`

```

vuongnguyenv@VNNOT02236:~$ kubectl get secret -n monitoring prom-operator-grafana \
onpath="{.data.admin-}" -o jsonpath="{.data.admin-}" | base64 --decode
prom-operator-vuongnguyenv@VNNOT02236:~$ kubectl get secret -n monitoring prom-operator-grafana \
> -o jsonpath="{.data.admin-}" | base64 --decode
prom-operator-vuongnguyenv@VNNOT02236:~$ kubectl get svc --all-namespaces

```

- After login successfully, visualization data via Dashboard

